

1 Introduction: The Problem of Cannons

In the 16th century, several scientists tried to find accurate models for how a cannonball flies.

Cannons were invented in China in the 13th century. From there, they spread to Europe, India, and Africa. At first, cannons were imprecise. If one shot went to the right, the next shot might go to the left, etc. It wasn't worth aiming those cannons carefully, since the shots would still go all over.

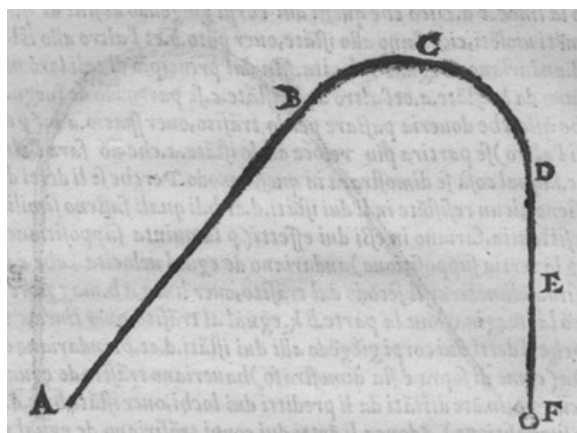
As cannons grew more accurate, aim started to matter more. Soldiers wanted to know things like how the angle at which they fired the cannon would affect how far the cannonball went. That's why scientists were trying to make accurate models.

The path a cannonball follows is called its *trajectory*. The early models of cannonball flight were the beginning of *ballistics*, a field applied today to sports, crime scene investigations, safety testing, and military devices. It can even be used to study cooking - a pizza crust tossed in the air has a lot in common with a cannonball!

Here's one 16th century model of a cannonball trajectory. The cannonball goes from right to left:



Below is a simplified version. (This one goes from left to right.)



This model was created by Niccoló Tartaglia. In Tartaglia's model, the trajectory has three different pieces:

1. a straight line coming out of the canon (A to B).
2. a portion of a circle (B to D).
3. a vertical drop to the ground (D to F).

There were special names for each stage of Tartaglia's model, such as "violent motion" for stage 1 and "natural motion" for stage 3, but those details won't matter to us. The key idea is that Tartaglia (and others of his time and earlier) thought of the cannonball as performing one type of motion at a time, and switching between different types of motion when some new influence took over. For example, they might say that the influence from the cannon would make the cannonball go in a straight line until that influence started to run out. Then after a transition period (the circle) the "natural motion" would take over, making the cannonball fall straight down. This type of thinking about motion goes back to Aristotle in ancient Greece.

Tartaglia's model isn't right. A cannonball doesn't move in a straight line at the beginning of its trajectory and it doesn't fall straight down at the end. It doesn't go in a circle in the middle, either! However, the model is close enough to the real motion of cannonballs that it gave some useful advice to soldiers.

Despite having some usefulness, Tartaglia's model, with its transitions between different types of motion, was conceptually far away from the improved models that were to come. Our goal today is to realize that:

1. A cannonball in flight is doing the same sort of motion at all times, not switching between different stages.
2. The motion is best understood by breaking it into two independent parts - horizontal and vertical - that happen simultaneously.

Cannonballs are far from the only object that gets hurled through the air. Animals jumping, balls kicked or thrown, or even rocks shot out of a volcanic eruption are all examples of the same sort of motion. These are all objects that leave the ground and, once in the air, aren't pushed or pulled by any significant influence other than Earth's gravity. These types of objects are called *projectiles*, so this week we are studying *projectile motion*.

Exercises

- 1.1. Which of the following would be well-modeled using projectile motion models?
- (A) a pole vaulter on their way up over the pole vault bar (You can search YouTube for "pole vault" if you need to know what this looks like.)
 - (B) a rocket constantly shooting fuel as it flies
 - (C) rice grains flying through the air as they are tossed in a wok
 - (D) water coming out of a drinking fountain
 - (E) a child swinging back and forth on a swing
 - (F) the same child after they jump off the swing in mid air

2 Watching a Projectile's Path

We can visualize the path of a projectile by taking a series of photographs as the projectile travels through the air, then superimposing the photos to see the location of the projectile at different times. Here are a few examples.



A man jumping off a diving board



A snowboarder flying off a ramp



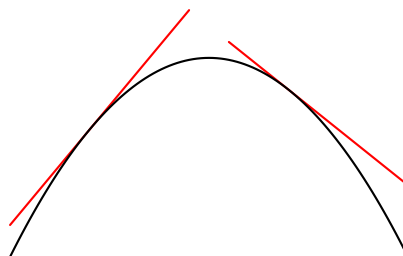
A basketball shot



A volcanic eruption. Each individual rock ejected is a projectile.

The volcano picture is in a different style than the others. It's a long-exposure photo, showing one particularly clear trajectory of a glowing projectile rock in the top right. This picture looks more like a solid line representing the trajectory. We can represent a trajectory either with dots, or with a solid line.

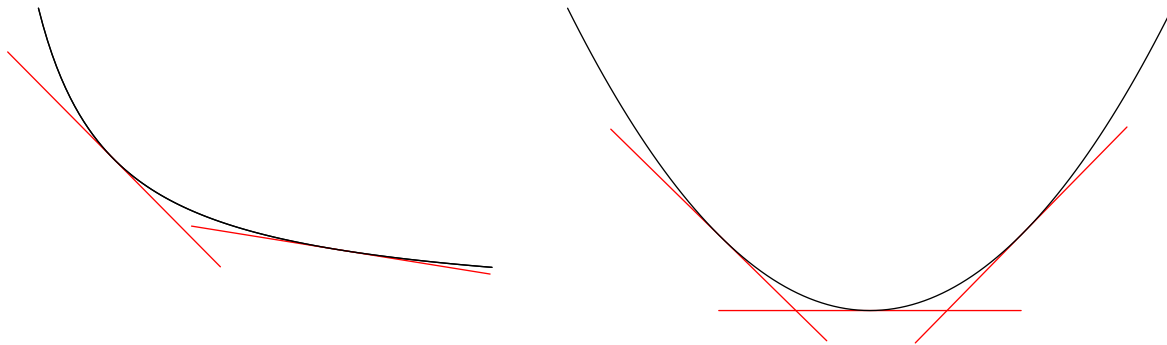
However we represent them, all the motions are quite similar! Some are steeper and some are shallower. Some go up and come down, while others have only the upward or only the downward portions. However, all the motions are curving downward all the time, meaning that if we draw a straight line along the path at any given point, the path lies below that line.



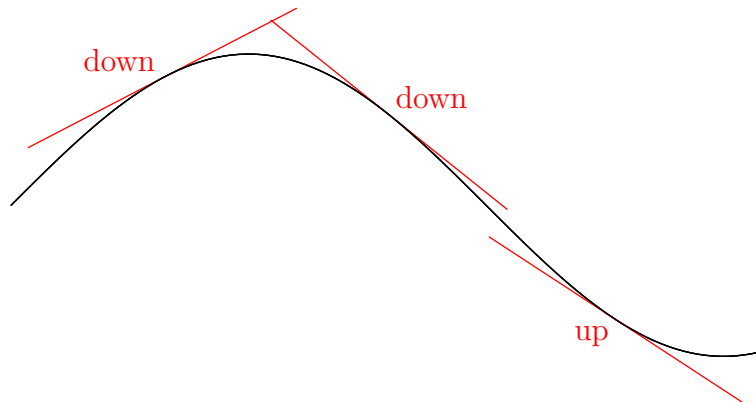
The path followed by a projectile is in black. The red lines are called *tangent lines*. The black curve bends down

below the tangent line, no matter which tangent line we look at. Whether projectiles are *moving* upward or downward, they always *curve* downward.

This point will be easier to understand with some contrast. Here are a couple of graphs which curve upward everywhere, since any tangent line is below the graph (everywhere except the point of tangency.)

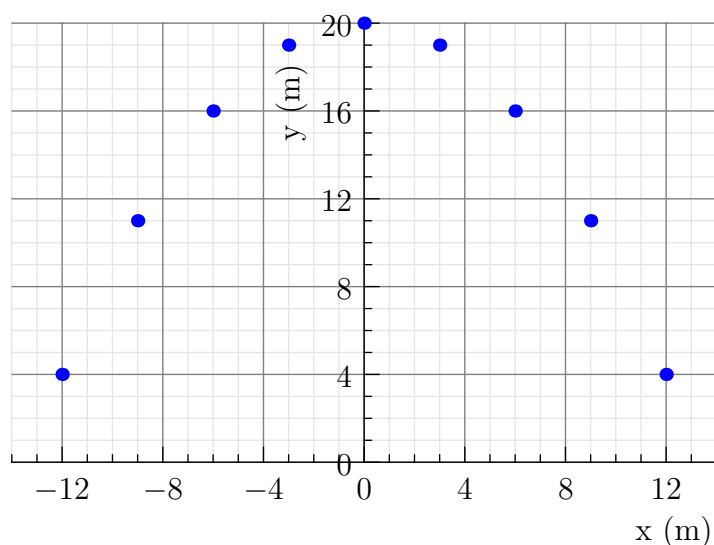


And many graphs curve down in some places and up in others.



Although there are many curves that always curve down, the idea that projectile trajectories always curve down still narrows down the set of possible trajectories.

To explore projectile trajectories further, below we have a dot diagram for an example projectile trajectory. We've also added coordinates: x for the horizontal direction and y for the vertical direction.



Exercises

- 2.1. Name something else that's a projectile.
- 2.2. The graphs we drew in the previous two weeks of class showed time on the horizontal axis. Why doesn't the dot diagram above show time on the horizontal axis? How does it represent time?
- 2.3. Where on the dot diagram are dots closer together? Where are they further apart?
- 2.4. Does your answer to the previous problem apply to the photographs as well?

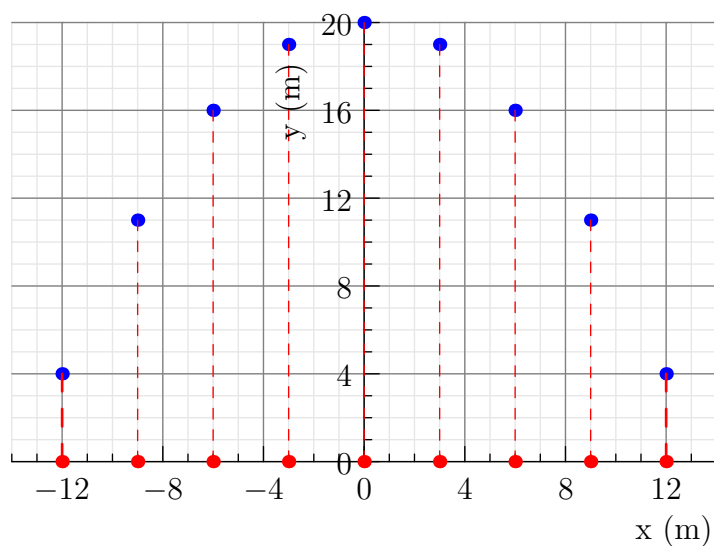
3 Two Motions Simultaneously

The dots are closer together near the top of an arc. Below is a photo of a bouncing tennis ball. The tennis ball pictures overlap with each other, but only near the top of the arc.



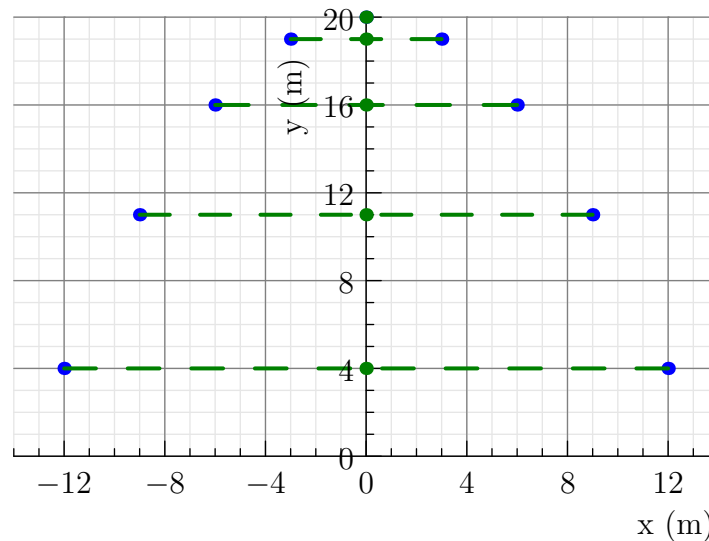
Since the distance between dots on a dot diagram decreases near the top of an arc, projectiles slow down as they rise. In what way? Do they change both their horizontal and vertical velocity components, or only one?

To find out, let's take a look at just the horizontal motion of the example projectile in the dot diagram at the end of section 2. We do this by putting red dots along the x axis directly underneath the original dots.



The red dots are evenly spaced, which we recognize as constant velocity motion.

We can also make a vertical dot diagram, which we'll do with green dots on the y axis.



Here are the green dots isolated, along with the spacing between them:



This is exactly the dot diagram we're used to for constant acceleration, and the acceleration works out to be $-g$.

Projectile motion is a combination of freefall in the vertical direction and constant velocity motion in the horizontal direction. These two motions happen simultaneously, and when we put them together, we get the trajectories we've been studying for jumpers, basketballs, rocks shot from volcanos, etc.

When we take a two-dimensional motion and isolate just one dimension of it, as we did in creating the red and

green dot diagrams, we call it *projection*. The horizontal motion is the projection of the trajectory onto the x axis, for example.

Each projection is called a *component* of the entire motion. The projection of the trajectory onto the horizontal axis (x axis) is called the *horizontal component*, or x *component*, of the motion. The other projection is the *vertical component*, or y *component*, of the motion. The full motion of the projectile is made from both components put together.

The first person to use this model for projectiles was Galileo. We'll call it "Galileo's model".

Galileo's model might seem more complicated than Tartaglia's because Galileo's model requires us to combine two components of motion to get the full motion. This turns out to be an advantage when solving problems, though. We can solve many problems using Galileo's model if we think about one component of the motion at a time. For example, if we want to know how high a projectile is at some time, we can look at just the y component of projectile motion. We then have a one-dimensional problem with constant acceleration, which is what we practiced last week. Finding the height this way is usually much easier than trying to do it with Tartaglia's model.

Galileo's model is also simpler than Tartaglia's model because Galileo didn't need any switching between different stages. In Galileo's model, the entire trajectory - rising and falling parts - consists of vertical acceleration of $-g$ and constant horizontal velocity.

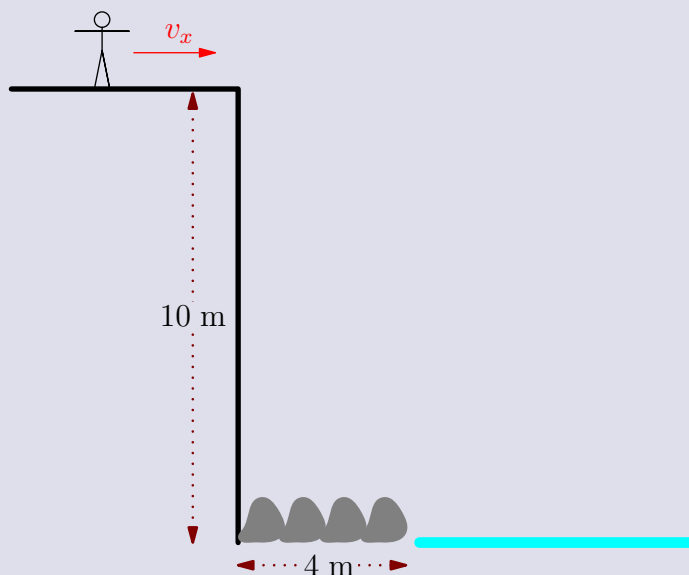
Galileo's model can reproduce all the trajectories we've seen, such as these two:



The tall, skinny trajectory of the volcano rock has a lower horizontal velocity than the initial vertical velocity. The rock doesn't move far in the horizontal direction, and the trajectory is narrow. The basketball has a higher horizontal

velocity than initial vertical velocity, so its trajectory is wide and short. Both projectiles have the same acceleration of $a_y = -g$ and $a_x = 0$.

Problem 3.1. A cliff diver wants to jump off a 10 m high cliff and land safely in the water below, but at the bottom there are rocks extending 4 m beyond the cliff. How fast v_x does the diver need to run towards the edge of the cliff to clear the rocks, assuming they run horizontally off the cliff? You can assume $g = 10 \text{ m/s}^2$.



Solution: We can think about the vertical and horizontal motions of the diver separately. First, we think about the vertical direction. The diver will fall 10 m vertically, and we can calculate how long that will take. When we studied free fall, we found that the distance Δy covered by an object in freefall in the vertical direction is

$$\Delta y = -\frac{1}{2}gt^2. \quad (3.1)$$

Setting $\Delta y = -h$, where h is the height of the cliff, and solving for t , we get

$$t = \sqrt{\frac{2h}{g}}. \quad (3.2)$$

Equation 3.2 gives the time to fall to the rocks, and it doesn't use v_x . That means that no matter how fast the person runs, they always take the same amount of time to fall. This is an example of how motion in the y direction is independent of motion in the x direction in Galileo's model.

Although many people find this feature of the model difficult to intuit at first, it matches experimental measurements, as we'll see in class. If the diver were to jump upward at some angle from the horizontal, that could affect their time of flight, but as long as their motion off the cliff is purely horizontal, that horizontal motion doesn't affect the vertical motion, and the vertical motion is what dictates how long it takes to fall.

Next we think about the horizontal direction. The horizontal component of velocity is constant. We can call it $v_{x;\min}$, so

$$d = v_x t \quad (3.3)$$

where d is the horizontal distance the diver travels before reaching the bottom of the cliff.

If we use t from equation 3.2 in equation 3.3, we get

$$d = v_x \sqrt{\frac{2h}{g}}. \quad (3.4)$$

Solving for v_x , the minimum running speed is

$$v_x = d \sqrt{\frac{g}{2h}}. \quad (3.5)$$

That's our proposed solution. This is our first projectile problem, so let's look at each of the proportionalities with v_x as a means of sensemaking. First, there's

$$v_x \propto d. \quad (3.6)$$

This makes sense - if we double how far out the rocks extend, we double how fast the diver has to run to clear them. The diver will take the same amount of time to hit the water no matter how fast they run off the rocks, so we get a linear proportionality between horizontal component of velocity and distance traveled horizontally.

Next, there's

$$v_x \propto \sqrt{\frac{1}{h}}. \quad (3.7)$$

This says that increasing the height of the cliff makes it easier for the diver to jump past the rocks, since it reduces v_x . However, doubling the cliff height doesn't cut the minimum running speed in half. Instead, it cuts the minimum running speed by a factor of $\sqrt{2}$.

Let's see where that factor comes from. If we double the height of the cliff, the diver needs to fall twice as far vertically before hitting the water. To make that happen, we multiply the time of the fall by $\sqrt{2}$, which also multiplies the average value of v_y by $\sqrt{2}$. Since $h = v_{y;\text{avg}} t$ and both $v_{y;\text{avg}}$ and t are being multiplied by $\sqrt{2}$, h is multiplied by 2.

Since doubling h multiplies t_{fall} by $\sqrt{2}$, it gives the diver $\sqrt{2}$ times as long to clear the rocks. Our minimum speed goes down by a factor of $\sqrt{2}$, just as the proportionality $v_x \propto \sqrt{\frac{1}{h}}$ shows.

Finally, there's

$$v_x \propto \sqrt{g}. \quad (3.8)$$

This says that the higher freefall acceleration is, the faster the diver needs to run to clear the rocks. Again, this is intuitively correct. There is a square root involved because doubling g would only cut the time it takes the diver to fall to the water by a factor of $\sqrt{2}$.

At last we can plug the values into the equation 3.5 for v_x .

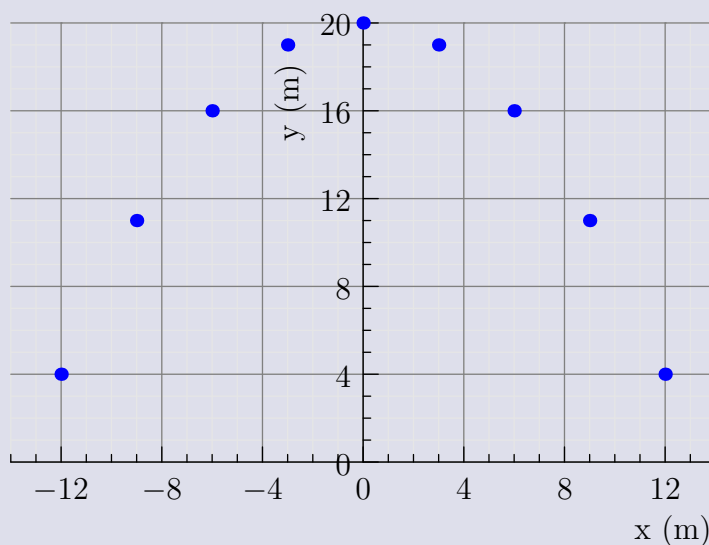
$$v_x = d\sqrt{\frac{g}{2h}} \quad (3.9)$$

$$= 4 \text{ m} \sqrt{\frac{10 \text{ m/s}^2}{2 \cdot 10 \text{ m}}} \quad (3.10)$$

$$= 2.8 \text{ m/s} \quad (3.11)$$

This is only a moderate running speed. Nonetheless, please don't test this calculation at home! $\square$

Problem 3.2. Find the initial horizontal and vertical velocities of the projectile in the dot diagram below. Take the leftmost dot as the initial position. Assume $g = 10 \text{ m/s}^2$.



Solution:

In the previous problem, we found that starting with zero vertical velocity, the time to fall a distance h is

$$t_{\text{fall}} = \sqrt{\frac{2h}{g}}. \quad (3.12)$$

The dot diagram says that the projectile fell from 20 m to 4 m, a fall of 16 m, so we could use equation 3.10 $h = 16 \text{ m}$ to calculate t_{fall} .

We don't need the numerical value t_{fall} , though. Instead, we want the vertical velocity of the projectile at the time of its takeoff.

If an object begins at rest and falls freely for a time t , its vertical velocity is $-gt$, so the vertical velocity at landing was

$$v_{y;\text{land}} = -gt_{\text{fall}} = -g\sqrt{\frac{2h}{g}} = -\sqrt{2hg}. \quad (3.13)$$

The rising portion of free fall motion is the same as the falling portion, just played backwards in time, so the vertical component of velocity at takeoff was

$$v_{y0} = \sqrt{2gh} = 17.9 \text{ m/s}. \quad (3.14)$$

(We often use a subscript of zero to indicate an quantity at the beginning of a scenario, such as the initial velocity. v_{y0} can be read “vee why naught” or “vee why zero”.)

Next, we want to find the horizontal component of the velocity. The projectile rose and then fell, so it was in the air twice as long as t_{fall} , or

$$t_{\text{flight}} = 2\sqrt{\frac{2h}{g}}. \quad (3.15)$$

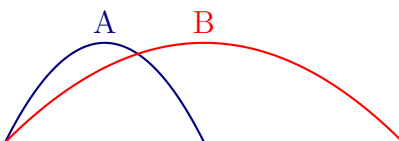
If we call the distance it traveled in that time Δx , then

$$v_x = \frac{\Delta x}{t_{\text{flight}}} = \frac{24 \text{ m}}{2\sqrt{\frac{2 \cdot 16 \text{ m}}{10 \text{ m/s}^2}}} = 6.7 \text{ m/s}. \quad (3.16)$$

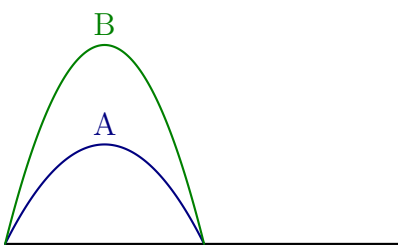
□

Exercises

3.1. Is it possible to tell from these two projectile paths which had the higher horizontal component of velocity and which had the higher initial vertical component of velocity? If so, say which one had the higher component in each direction.



3.2. Answer the same question about these two trajectories:



- 3.3. You're standing inside a train moving steadily down straight tracks at 40 m/s. You jump straight up and land 0.5 s later. How far would someone outside the train say that you moved forward during your jump? How far would someone inside the train say you moved?
- 3.4. You want to fire a cannonball perfectly horizontally, and fire it so fast it travels at least the first 100 m horizontally without falling at all. What's the minimum speed to accomplish this, or is it not possible?
- 3.5. You want to fire a cannonball *almost* horizontally, so that it travels 100 m horizontally while only rising and then falling 5 cm. What's the initial vertical velocity you should give the cannonball, and what's the horizontal velocity you should give it?
- 3.6. (optional) A cannonball is launched at 45° above the horizontal, so that the initial vertical and horizontal components of its velocity are equal. If the cannonball is to land 1000 m away from where it was fired, how high will it go, assuming negligible air resistance?

4 An equation for the trajectory

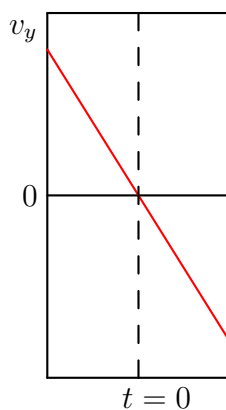
Problem 4.1. Find $y(x)$, the height as a function of horizontal position, for a projectile launched in terms of v_x and the maximum height of the trajectory, $y_{\max}$.

Solution: We'll set up our coordinates so that the horizontal position of the projectile is $x = 0$ at time $t = 0$. Then the horizontal position is

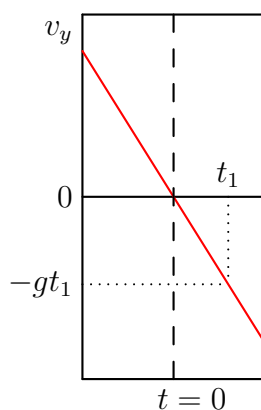
$$x(t) = v_x t. \quad (4.1)$$

Next, we'll analyze the vertical position as a function of time. Let's choose the zero of time so that the projectile reaches its maximum height right at that time. Since the projectile is at its maximum height, it is not moving upward or downward, and its vertical velocity is zero.

Remembering that the vertical motion has constant acceleration $a_y = -g$, the slope of a v_y versus time graph is $-g$, and the graph looks like this:



If we choose some time t_1 , the vertical component of velocity at that time is $v_y(t_1) = -gt_1$.



The average vertical component of velocity between time $t = 0$ and time t_1 is half way between the velocities at those two times, or

$$v_{\text{avg};0 \rightarrow t_1} = \frac{1}{2} \cdot (-gt_1). \quad (4.2)$$

The displacement of the projectile in the vertical direction is

$$\Delta y(t_1) = v_{\text{avg};0 \rightarrow t_1} \cdot t_1 = \left(\frac{-gt_1}{2} \right) \cdot t_1 = -\frac{gt_1^2}{2}. \quad (4.3)$$

This is true at any time t , so we can replace t_1 with t . Δy is the displacement from the maximum height of the projectile, so $y = y_{\text{max}} + \Delta y$, where y_{max} is the maximum height. Then we have

$$y = y_{\text{max}} - \frac{gt^2}{2}. \quad (4.4)$$

We now have two equations: one for $x(t)$ and one for $y(t)$.

Since we wanted to know just the shape of the trajectory, not the times it's at different places, we can eliminate t via algebra. We start by solving equation 4.1 for t .

$$t = \frac{x}{v_x}. \quad (4.5)$$

We can plug this into equation 4.4.

$$y = y_{\max} - \frac{g \left(\frac{x}{v_x} \right)^2}{2}. \quad (4.6)$$

Simplifying, we have an equation relating y and x ,

$$y = y_{\max} - \frac{g}{2v_x^2} x^2. \quad (4.7)$$

This is the equation for a *parabola*. This equation applies to any projectile we can model as having constant horizontal velocity and constant vertical acceleration $-g$, as long as the horizontal velocity is not zero.

Of course, the model only applies while the projectile is still in the air, so only for x values between the takeoff and landing points. We will usually say that projectile motion is modeled by a parabola, but it would be more accurate to say we model it with a portion of a parabola - a parabolic arc. $\square$

Exercises

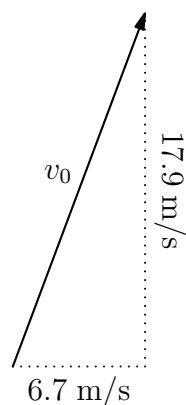
- 4.1. Could a parabola model motion where the y component of velocity is always positive? Explain your reasoning.
- 4.2. (Optional) Write the equation for $y(x)$ assuming the projectile reaches its maximum height at position $x = x_{\max}$ instead of $x = 0$.
- 4.3. (Optional) Write the equation for $y(x)$ assuming the projectile reaches its maximum height at time $t = t_{\max}$ instead of $t = 0$. (Forget about the $x_{\max}$ from the previous exercise.)
- 4.4. (Optional) Write the equation for $y(x)$ assuming the projectile is launched from the point $(0, 0)$. Instead of $y_{\max}$, use the initial y velocity v_{y0} .

5 Angles and Combining Velocities

Problem 5.1. Find the initial speed of the projectile from the dot diagram above.

Solution: The horizontal component of speed was $v_x = 6.7$ m/s and the vertical component was $v_{y0} = 17.9$ m/s. (We haven't bothered adding a 0 subscript to v_x because it is constant throughout the trajectory.)

We can draw the initial velocity as an arrow.



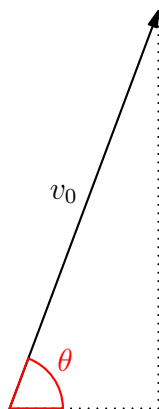
The arrow points more upward than horizontal because the vertical component of velocity is significantly larger. We've also shown the horizontal and vertical components of the initial velocity as dotted lines.

The initial speed v_0 is the length of the hypotenuse of the triangle, which we can find using the Pythagorean theorem. It is

$$v_0 = \sqrt{v_x^2 + v_{y0}^2} = 19.1 \text{ m/s}. \quad (5.1)$$

□

A cannon can be shot at different angles. We usually measure the angle that the cannon barrel points above the horizontal and call it θ .



Given the initial speed and the angle a projectile is fired from, it is possible to find the initial horizontal and vertical components of velocity. This week, we'll only do that for angles of 30° , 45° , or 60° , which we can handle more easily than other angles.

6 Air Resistance and Other Complications

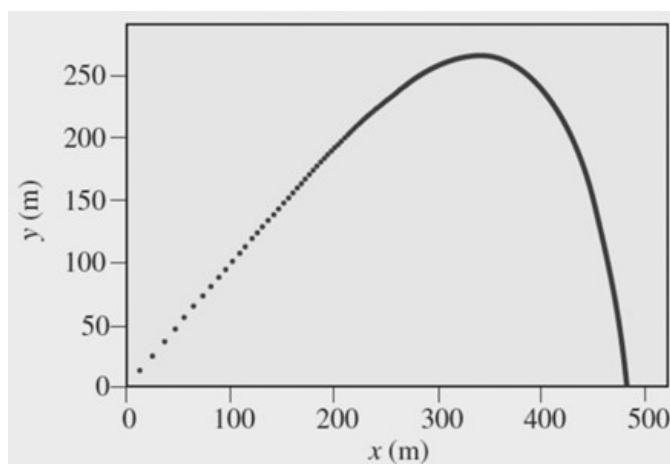
Galileo did not really solve the problem of what a cannonball's trajectory looks like. His model didn't include the effect of air resistance on the cannonball as it flies.

If you're tossing an apple back and forth with a friend from a few meters apart, air resistance doesn't affect the trajectory of the apple significantly. The air doesn't push on the apple enough to move it off course much.

A cannonball is fired much faster than an apple, though, and travels much further. The effects of air pushing on the cannonball add up and create large errors in Galileo's model. He understood this and stated that his model was only for situations where the air could be ignored, but he wasn't able to create accurate models for when the air is important.

Today, we have models that can include air drag, as well as other influences on projectiles, such as the Coriolis acceleration, or even the fact that free fall acceleration changes over the trajectory, since it's lower the further we go from the center of Earth.

Here's a dot diagram for a model that starts with Galileo's model, but adds in air drag.



It winds up looking a lot like Tartaglia's model!

In any given situation, we may want to know whether air drag is important. A heuristic is that air drag becomes important when a solid object moves several hundred times its own diameter through the air.

For example, an apple has a diameter of around 10 cm, so when the apple is thrown for thousands of centimeters, or tens of meters, air drag starts to have a significant influence.

This is only a rough guideline. The effects of air drag are still there when an apple is thrown two or three meters, but they're small.

How small the effects of air drag have to be before we can ignore them also depends on how precise we want to be. If we only want to know what the trajectory of an apple will look like to within 5 cm, we can get away with much more air drag than if we want to know the trajectory all the way down to 0.1 mm of uncertainty.

Additionally, when projectiles spin, the effects of air become more complicated, and can provide "lift" that pushes the projectile upward, or can push it to the side. This can become important in sports where balls curve while in the air.

Hollow objects are easier to push around, and we'll see the effects of air drag over shorter trajectories with them. You can see significant effects of air drag volleying a beach ball back and forth, for example, and the effects of air are very large on a soap bubble.

Very dense things, like metal cannonballs, will be affected less by air than less dense things, like fruit or wooden balls. The distance a projectile can be shot with a given amount of deflection due to air is roughly proportional to the density of the projectile, for fixed projectile diameter. If we double the density of a projectile, it can go twice as far with the same amount of deflection from air, approximately.

This week, we'll focus on using Galileo's model with no air drag. Using a model that is wrong because it ignores air drag isn't necessarily bad. However, it means our calculations have limitations. It's similar to the situation from the first two weeks of the course: objects don't move with truly constant velocity, but a constant velocity model has small enough error to work well in some applications. An elevator doesn't truly have constant acceleration, but a constant acceleration model again works pretty well there. Galileo's projectile motion model is never perfect, either, but in many cases it's good enough that we don't need a more detailed model.

As we learn more physics, we'll be able to build more complicated models that take new effects into account, but in the real world, there's always *something* you're leaving out of your analysis. Scientists have a saying, "All models are wrong, but some are useful."

Exercises

- 6.1. A cannonball in Galileo's time was about the same size as an apple, but about five times as dense. Would there be significant deflection to a cannonball shot 500 m?
- 6.2. Besides effects from air, the Coriolis acceleration, and changing freefall acceleration, name one other thing, however small, that you think could affect the trajectory of a projectile.

7 References and Notes

We haven't tried to portray the ideas about projectile motion from before Galileo in perfect accuracy. Western modeling of motion traces back largely to Aristotle in ancient Greece, and progressed in bits and pieces after Aristotle. Galileo didn't find the first accurate model out of nothing! Tartaglia's model is from 1531. Although his model was not as accurate or sophisticated as modern models, he understood several subtleties about the motion, including that air resistance was a complicating factor, and that if we build a model that neglects air resistance, the model will be simpler, but less accurate than one that includes it. He also didn't have a strict belief in a sharp transition from one type of motion to another, and didn't think stage 1 was purely straight line motion, but instead that gravity had an influence even during that stage, and the motion was just very close to a straight line. For more detail on Tartaglia's model, see

Valleriani, Matteo (2013). "Tartaglia's Work on Theoretical Ballistics". In: *Metallurgy, Ballistics and Epistemic Instruments: The Nova scientia of Nicolò Tartaglia – A New Edition*. Berlin: Max-Planck-Gesellschaft zur Förderung der Wissenschaften.

8 Credits

early model of cannonball, scene: Rocca, P. and Riggi, Francesco. (2009). Projectile motion with a drag force: Were the Medievals right after all?. *Physics Education*. 44. 10.1088/0031-9120/44/4/009.

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9 Answers to exercises

1.1: (C), (D), and (F). The pole vaulter is not because they are pushing off their pole. The rocket is not because the fuel pushes the rocket forward. The child on the swing is not because the swing pushes on them.

2.1: Anything thrown through the air is fine.

2.2: Because we need to model two dimensions of motion through space. Time is represented by counting the dots.

2.3: Closer higher up, further apart lower down.

2.4: Yes.

3.1: Yes. B had higher horizontal velocity, and they had the same initial vertical velocity. We can tell the initial vertical velocities based on how high they go. Since they go the same height, they had the same initial vertical velocity. This also means they were in the air for the same amount of time, so B had to be moving horizontally at higher speed.

3.2: B had the higher initial vertical velocity since it goes higher. A had the higher horizontal velocity because it goes the same distance in less time.

3.3: 20 m and 0 m.

3.4: Not possible; it will have acceleration $-g$ downward from the moment it's fired.

3.5: $v_{y0} = 1 \text{ m/s}$ to get a rise of 5 cm. $v_x = 500 \text{ m/s}$ to travel that far in the 0.2 s it takes to rise and then fall 5 cm.

3.6: 250 m. When the cannonball is at its highest elevation, it has traveled half its range, or 500 m. The vertical velocity began equal to the horizontal velocity, but steadily reduced to zero. On average, the vertical velocity has been half the horizontal velocity, so the cannonball has risen half of 500 m, or 250 m.

4.1: Yes, as long as we restrict the x values to all be negative, we could use the parabola in this section.

4.2: $y = y_{\max} - \frac{g}{2v_x^2}(x - x_{\max})^2$

$$4.3: y = y_{\max} - \frac{g}{2v_x^2}(x - v_x t_{\max})^2$$

$$4.4: y = \frac{v_{y0}}{v_x}x - \frac{g}{2v_x^2}x^2.$$

6.1: Yes. We estimated that air influences start to become significant when throwing an apple a few tens of meters. Moving to a cannonball, we're multiplying the density by only five, while multiplying the range by 50. At this long range, air resistance will be a large factor.

6.2: Any answer is okay. Light from the sun could push on the projectile. Pieces of the projectile could fall off. The projectile might hit something besides air (like an insect) while it was flying. Our model doesn't take into account the theory of relativity! You can probably come up with more examples.